

Chapter 31 — Tuberculosis and Parasitic Infections of the Genitourinary Tract — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **Urogenital (UG) tuberculosis (TB) gold standard:** urine acid-fast bacilli (AFB) culture, 3–5 first-void samples
- **Sterile pyuria ± haematuria** → think TB (> 90% in endemic series)
- **Kidney 80%** of UG TB; **epididymis** = commonest genital site
- **Standard UG TB:** 6 months — 2 months isoniazid (INH) + rifampicin + pyrazinamide + ethambutol (RIPE) → 4 months INH + rifampicin
- **Operate 4–6 weeks** after starting anti-TB drugs
- **Tuberculin skin test (TST) / interferon-γ release assay (IGRA)** = infection only; cannot diagnose or exclude active TB
- **Schistosoma haematobium:** eggs in urine = gold standard; collect 09:00–15:00
- **Praziquantel (PZQ)** 40 mg/kg once or 2 × 20 mg/kg — only World Health Organization (WHO)-recommended drug
- **Schistosomal bladder cancer:** squamous cell carcinoma (SCC) 60–90%, onset 40–50 years
- **Genital filariasis** = *Wuchereria bancrofti*; diethylcarbamazine (DEC) = drug of choice

UROGENITAL TUBERCULOSIS

- **Cut-offs:** ¼ of world latently infected; UG TB in 15–20% (developing) vs 2–10% (developed) of pulmonary TB
- **Primary progression** < 5%; **lifetime reactivation** 5–15%; UG latency up to 46 years
- **Male : female** ≈ 2 : 1; bacille Calmette–Guérin (BCG) instillation → UG TB ≈ 1%
- **Constitutional symptoms** < 20%; **renal colic** < 10%
- **Autonephrectomy** up to ⅓; **end-stage renal failure** ≈ 7%
- **Epididymis** 10–55% (bilateral ⅓); **prostate** 22–49%; **urethra** 1.9–4.5%
- **Bladder** < 100 mL → symptomatic; **thimble bladder** < 20 mL
- **Gold standard:** urine AFB culture; sensitivity 80–90% ideal vs ≈ 10% real-world
- **Ziehl–Neelsen urine stain** < 50% sensitive
- **Culture time:** Löwenstein–Jensen 4–6 weeks; Middlebrook 7H10 ≈ 1 week sooner; Mycobacteria Growth Indicator Tube (MGIT, liquid) ≈ 10 days
- **Always one solid medium** alongside liquid culture
- **Nucleic acid amplification test (NAAT):** 1–2 days; adjunct only; cannot monitor treatment
- **GeneXpert rifampicin resistance** = surrogate for multidrug-resistant TB (MDR-TB) (> 90% also INH-resistant)
- **Urine lipoarabinomannan (LAM) lateral-flow** — **WHO 2019:** people with HIV + symptoms, advanced HIV, CD4 < 200
- **TST, read 48–72 h:** ≥ 5 mm contacts / immunosuppressed / fibrotic chest film; ≥ 10 mm high-risk groups; ≥ 15 mm no risk
- **IGRA preferred** ≥ 5 years from endemic country; unaffected by BCG
- **Sensitivity:** T-SPOT.TB > TST > QuantiFERON (third generation)
- **Chest radiograph for every patient** — coexistent pulmonary TB ≤ 20%
- **Imaging now:** computed tomography (CT) urography; **former gold standard** intravenous urography (IVU)
- **Commonest IVU finding:** obstruction; stricture commonest at distal ureter / vesicoureteric junction
- **Putty / cement kidney** (lobar calcification) = pathognomonic of end stage
- **Kerr's kink:** hiked-up pelvis, sharp pelviureteric junction (PUJ) angulation
- **Signs:** phantom calyx, moth-eaten calyx, pipestem / beaded-corkscrew ureter, golf-hole orifice
- **Children / pregnancy:** ultrasound (US) or magnetic resonance imaging (MRI); US tracks hydronephrosis on treatment
- **First line:** RIPE by directly observed therapy
- **Continuation:** daily or 3× weekly — never twice weekly
- **Doses:** INH 5 mg/kg (300 mg); rifampicin 10 mg/kg (600 mg); pyridoxine 25–50 mg
- **Ethambutol:** only non-hepatotoxic first-line drug → check acuity and red-green vision
- **Complicated UG TB:** 12–24 months, or ≥ 6 months after surgery
- **Pregnancy:** INH + rifampicin + ethambutol + pyridoxine, 9 months
- **HIV not on effective antiretrovirals** → 9 months, daily dosing
- **Transplant:** rifamycin-free regimen ≥ 18 months; mortality up to 36%
- **Rifabutin** = rifamycin of choice with antiretrovirals
- **Pyrazinamide in pregnancy:** omitted in United States; WHO recommends
- **MDR** = INH + rifampicin resistance; cure 50–60% vs 94–97% susceptible; **XDR** 30–50%
- **Pre-XDR:** MDR + fluoroquinolone or injectable aminoglycoside
- **BPAL** (bedaquiline + pretomanid + linezolid) 6–9 months; not approved for extrapulmonary TB
- **Operate if:** uraemia / sepsis → retrograde stent first (succeeds 41%) → antegrade → nephrostomy
- **Operate if:** non-functioning kidney + recalcitrant TB (relapse < 1%) or resistant hypertension (65% improve)
- **Operate if:** stricture no better after 6 weeks of drugs → open repair
- **Nephrectomy:** laparoscopic preferred; ligate renal artery and vein separately
- **Nephrostomy removed without fixing cause** → cutaneous TB fistula
- **Gap bridging:** mobilisation 2–3 cm; psoas hitch ≤ 5 cm; Boari flap 10–15 cm; beyond → ileal ureter
- **Bladder:** < 100 mL → augmentation (ileocaecum / sigmoid); < 20 mL → orthotopic substitution
- **Epididymis only** → epididymectomy; ligate vas at external ring
- **Do not operate:** non-functioning kidney without recalcitrant TB or hypertension
- **HIV** → 15–20× risk of active TB; UG TB less fibrotic, more lethal
- **Relapse:** UG 6.3–22% vs pulmonary 2–6%; mean 5.3 years → follow up 10 years

PARASITIC INFECTIONS OF THE UROGENITAL TRACT

Schistosomiasis

- **S. haematobium:** *Bulinus* snail; vesical venous plexus; egg ≈ 140 µm, terminal spine
- **Bilharz (1852):** worms in mesenteric veins linked to urinary eggs
- **Cercariae penetrate intact skin;** none shed < 15 °C or > 35 °C
- **Egg-laying** 8–12 weeks; **antibodies** 4–6 weeks
- **Single exposure (visitors):** worms die within 3–5 years
- **Katayama fever** 2–8 weeks; mainly *S. japonicum*; add steroids
- **Obstruction:** juxtavesical ureter commonest (20–60%); vesicoureteric reflux in 25%
- **Tonic hydroureter** (hypertrophied) reversible vs **atonic** (atrophic)
- **Tubular concentrating function** lost before glomerular
- **Symmers fibrosis:** pipestem periportal fibrosis from eggs
- **Salmonella in worm tegument** → treat both infections
- **Contracted bladder** ≤ 50 mL; trigone spared
- **40–60%** of obstructive uropathy presents at inactive end stage
- **Cancer:** International Agency for Research on Cancer (IARC) Group 1; SCC 60–90%, adenocarcinoma 5–15%; posterior wall
- **Urine peak noon;** rectal snip before bladder biopsy; squash preparation > histology
- **Serology** (FAST-ELISA + Western blot) > 90%; stays positive after cure
- **Active infection:** eggs or circulating anodic / cathodic antigen (CAA / CCA); **most sensitive:** polymerase chain reaction (PCR)
- **Plain film:** calcified bladder = "fetal head in pelvis"; **earliest IVU sign:** striations
- **PZQ** 2 × 20 mg/kg 6–8 h apart; efficacy 60–90%; misses schistosomulae → repeat in weeks
- **Artemether / artesunate:** 90–97% prophylaxis; not in malaria co-endemic areas
- **Active polypoid obstruction** → **PZQ**; renal function recovers in 1–2 months
- **Operate if:** severe haemorrhage; failed medical therapy; contracted bladder; chronic ulcer → partial cystectomy
- **Residual stenosis:** Politano–Leadbetter reimplant variants; dilatation restenoses
- **Aswan Low Dam (1930s):** prevalence < 11% → > 75%
- **Egyptian mass treatment:** fewer bladder cancers; SCC → transitional cell shift
- **WHA65.21 (May 2012):** morbidity control → elimination; eliminated in 10 countries
- **Kill cercariae:** water > 125 °F (≈ 52 °C) for 5 minutes, or stand > 2 days

Filariasis

- **W. bancrofti** > 90%; nocturnal microfilaraemia; mosquito vector
- **Bancroftian:** whole limb + genitals; **brugian:** below knee, genitals spared

- **Filarial lymphangitis retrograde** (peripheral spread)
- **Filarial dance sign** on US (≥ 80%); midnight Giemsa smear
- **Antigen test** for *W. bancrofti* only; burned-out cases antibody-positive
- **DEC** 2 mg/kg three times daily; 1 day = 12 days; tropical pulmonary eosinophilia 2–3 weeks
- **Albendazole** 400 mg twice daily 21 days: micro- + macrofilaricidal; **ivermectin** mainly microfilaricidal
- **Tropical pulmonary eosinophilia:** nocturnal wheeze, high IgE, microfilaraemia absent
- **Doxycycline** 200 mg 4–8 weeks: anti-*Wolbachia*, sterilises adults
- **Hydrocele:** excise intact sac
- **Lymph scrotum:** ruptured scrotal lymphatic vesicles, whitish discharge
- **Chyluria** precedes genital elephantiasis
- **Mass drug administration:** albendazole + ivermectin (Africa) / + DEC elsewhere
- **Onchocerciasis:** ivermectin 150 µg/kg every 6–12 months; "hanging groin"
- **Loa loa:** DEC; encephalopathy risk > 2500–8000 microfilariae/mL; no *Wolbachia* → no doxycycline

Other parasites

- **Hydatid kidney** 2–3% (third site): albendazole 400 mg twice daily 1–6 months; no puncture-aspiration-injection-re-aspiration (PAIR)
- **Pinworm:** albendazole 400 mg or mebendazole 100 mg once; repeat at 2 weeks; treat contacts
- **Amoebiasis:** metronidazole / tinidazole, then luminal agent (paromomycin / iodoquinol)

CLASSIC TRAPS

- **Trap:** BCG / *M. bovis* = *M. tuberculosis* — **Actually:** innately pyrazinamide-resistant
- **Trap:** quinolone-responsive prostatitis excludes TB — **Actually:** relapse after short course suggests TB
- **Trap:** negative NAAT excludes UG TB — **Actually:** adjunct only; culture needed
- **Trap:** ureteric calcification = TB — **Actually:** TB intraluminal; schistosomiasis intramural, dilated ureter
- **Trap:** healing relieves obstruction — **Actually:** fibrosis worsens it; monitor with US
- **Trap:** 4-month rifapentine regimen for UG TB — **Actually:** not for extrapulmonary TB
- **Trap:** schistosomal serology confirms cure — **Actually:** stays positive; use eggs or CAA / CCA
- **Trap:** egg-negative schistosomiasis = mild — **Actually:** inactive stage has worst kidneys
- **Trap:** DEC for any filaria — **Actually:** contraindicated in onchocerciasis and heavy *Loa*
- **Trap:** funiculopididymitis = tumour → orchidectomy — **Actually:** filarial inflammation; preserve testis